Assessment Menu: Teacher Guide

The Volume Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
AREA	Area of composite figures	A1, A2, E1, S1
PRISM	Surface area and volume of prisms and cylinders	A3, E2, E3, S2
PYRAMID	Pyramids, cones, and spheres	A4, A5, E4, S3
MODEL	Unit conversions and modeling	A6, E5, E6, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 AREA	Find the area of a trapezoid with parallel sides 8 cm and 14 cm and height 5 cm.	**Answer:** 55 cm² $A = \frac{1}{2}(b_1 + b_2)h$ $A = \frac{1}{2}(8 + 14)(5) = \frac{1}{2}(22)(5) = 55$ cm²	2 pts: Correct trapezoid formula 1 pts: Substitutes 8, 14 and 5 2 pts: Area 55 cm² (square units)
A2 AREA	An L-shaped patio is a 10 m by 6 m rectangle with a 4 m by 3 m rectangular corner removed. Find its area.	**Answer:** 48 m² Whole rectangle: 10 × 6 = 60 m² Removed corner: 4 × 3 = 12 m² Patio: 60 − 12 = 48 m²	2 pts: Whole rectangle 60 m² 1 pts: Removed corner 12 m² 2 pts: Subtracts: 48 m²
A3 PRISM	A rectangular prism is 5 in by 4 in by 3 in. Find its volume and its surface area.	**Answer:** V = 60 in³; SA = 94 in² V = 5 × 4 × 3 = 60 in³ SA = 2(5·4) + 2(5·3) + 2(4·3) = 40 + 30 + 24 = 94 in²	2 pts: Volume 60 in³ 3 pts: Surface area 94 in² (all six faces)
A4 PYRAMID	Find the volume of a cone with radius 3 cm and height 7 cm. Leave your answer in terms of π.	**Answer:** 21π cm³ $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(3^2)(7)$ $= \frac{1}{3}\pi(63) = 21\pi$ cm³	2 pts: Uses V = one third πr²h 3 pts: Volume 21π cm³
A5 PYRAMID	A sphere has radius 6 in. Find its volume and its surface area, in terms of π.	**Answer:** V = 288π in³; SA = 144π in² $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(216) = 288\pi$ in³ SA = 4πr² = 4π(36) = 144π in²	3 pts: Volume 288π in³ 2 pts: Surface area 144π in²
A6 MODEL	A room is 4 yd by 5 yd. Find its area in square feet. (1 yd = 3 ft)	**Answer:** 180 ft² Convert each side: 4 yd = 12 ft and 5 yd = 15 ft Area = 12 × 15 = 180 ft²	2 pts: Converts the sides to 12 ft and 15 ft 1 pts: Multiplies the sides 2 pts: Area 180 ft²

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 AREA	A window is a rectangle 4 ft wide and 6 ft tall, topped by a semicircle whose diameter is the rectangle's width. Find the total area in terms of π, then to the nearest tenth of a square foot.	**Answer:** $(24 + 2\pi)$ ft² ≈ 30.3 ft² Rectangle: 4 × 6 = 24 ft² Semicircle: radius 2, so area = $\frac{1}{2}$ π(2²) = 2π ft² Total = 24 + 2π ≈ 24 + 6.28 = 30.3 ft²	2 pts: Rectangle 24 ft² 2 pts: Radius is 2 (half of 4) 3 pts: Semicircle area 2π ft² 1 pts: Adds the two areas 2 pts: Decimal 30.3 ft²
E2 PRISM	A triangular prism has a right-triangle base with legs 6 cm and 8 cm and hypotenuse 10 cm. The prism is 12 cm long. Find its volume and its total surface area.	**Answer:** V = 288 cm³; SA = 336 cm² Base area = $\frac{1}{2}(6)(8)$ = 24 cm² V = 24 × 12 = 288 cm³ Two bases: 2(24) = 48 cm² Three rectangles: (6 + 8 + 10) × 12 = 288 cm² SA = 48 + 288 = 336 cm²	2 pts: Base area 24 cm² 2 pts: Volume 288 cm³ 2 pts: Two bases 48 cm² 2 pts: Lateral area (6 + 8 + 10) × 12 = 288 cm² 2 pts: Surface area 336 cm²
E3 PRISM	A cylinder has volume 150π cm³ and height 6 cm. Find its radius, then its total surface area, in terms of π.	**Answer:** r = 5 cm; SA = 110π cm² 150π = πr²(6), so r² = 25 and r = 5 cm Bases: 2πr² = 2π(25) = 50π cm² Side: 2πrh = 2π(5)(6) = 60π cm² SA = 50π + 60π = 110π cm²	3 pts: Sets up 150π = πr²(6) 2 pts: r = 5 cm 3 pts: Both parts of the surface area (50π and 60π) 2 pts: SA 110π cm²
E4 PYRAMID	A square pyramid has base edge 10 cm and slant height 13 cm. **(a)** Find its surface area. **(b)** Use the height, half the base, and the slant height to find the pyramid's height, then its volume.	**Answer:** (a) 360 cm² (b) h = 12 cm, V = 400 cm³ (a) Base: 10² = 100. Each triangle face: $\frac{1}{2}(10)(13)$ = 65, so 4 faces = 260. SA = 360 cm². (b) Right triangle: 5² + h² = 13², so h² = 144 and h = 12 cm. V = $\frac{1}{3}(100)(12)$ = 400 cm³	2 pts: Base area 100 cm² 2 pts: Four triangular faces 260 cm² 1 pts: SA 360 cm² 2 pts: Height from 5² + h² = 13²; h = 12 3 pts: Volume 400 cm³ using one third
E5 MODEL	A box is 2 ft by 3 ft by 1 ft. Kim says: "Its volume is 6 ft³, which is 6 × 12 = 72 in³." Find Kim's mistake and give the correct volume in cubic inches.	**Answer:** Cubic units need 12³ = 1728, not 12. V = 10,368 in³. Volume is 2 × 3 × 1 = 6 ft³ (correct). Kim's mistake: 1 ft³ is 12 × 12 × 12 = 1728 in³, not 12 in³ (three dimensions each convert). 6 × 1728 = 10,368 in³ Check: 24 in × 36 in × 12 in = 10,368 in³	4 pts: Identifies mistake: a cubic foot is 12³ = 1728 cubic inches 3 pts: Multiplies 6 × 1728 3 pts: Correct volume 10,368 in³
E6 MODEL	A tank is 50 cm long, 30 cm wide and 40 cm tall, filled with water to 5 cm below the top. (1 L = 1000 cm³) **(a)** How many liters of water are in it? **(b)** If 12 L is removed, how far does the water level drop?	**Answer:** (a) 52.5 L (b) 8 cm (a) Water depth = 40 − 5 = 35 cm. Volume = 50 × 30 × 35 = 52,500 cm³ = 52.5 L. (b) 12 L = 12,000 cm³. The base area is 50 × 30 = 1500 cm². Drop = $\frac{12{,}000}{1500}$ = 8 cm	2 pts: Water depth 35 cm 2 pts: Volume 52,500 cm³ 1 pts: 52.5 L 2 pts: 12 L = 12,000 cm³ 3 pts: Drop = 12,000 divided by 1500 = 8 cm

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 AREA	A park is a 40 m by 30 m rectangle. A 12 m by 12 m square playground and a semicircular pond with diameter 20 m lie inside it, not overlapping. **(a)** Find the grass area in terms of π, then to the nearest m². **(b)** A bag of seed covers 150 m². How many bags are needed? **(c)** Rosa says doubling the pond’s diameter doubles its area. Is she right? Explain.	**Answer:** (a) $(1056 - 50\pi)$ m² ≈ 899 m² (b) 6 bags (c) No; the area quadruples. (a) Park $40 \times 30 = 1200$; playground $12^2 = 144$; pond $\frac{1}{2}\pi(10^2) = 50\pi$. Grass $= 1200 - 144 - 50\pi = 1056 - 50\pi \approx 899$ m² (b) $\frac{899}{150} \approx 5.99$. Partial bags can’t be bought, so round up: 6 bags. (c) No. Area uses r^2, so doubling the diameter multiplies the area by $2^2 = 4$. Example: radius 10 gives 50π; radius 20 gives 200π, which is 4 times as much.	4 pts: (a) Park 1200 and playground 144 4 pts: (a) Pond area 50π 2 pts: (a) $1056 - 50\pi \approx 899$ m² 4 pts: (b) About 5.99 rounded up to 6 bags, with reason 6 pts: (c) Says no, explains area depends on r^2 so it quadruples (3 pts), supports with numbers (3 pts)
S2 PRISM	Can A has radius 4 cm and height 9 cm. Can B has radius 6 cm and holds the same volume. **(a)** Find Can A’s volume. **(b)** Find Can B’s height. **(c)** Find each can’s label area (side only). **(d)** Which can uses less label material, and by how much? Use π in answers.	**Answer:** (a) 144π cm³ (b) 4 cm (c) A: 72π cm², B: 48π cm² (d) Can B, by 24π cm² (a) $V = \pi(4^2)(9) = 144\pi$ cm³ (b) $\pi(6^2)h = 144\pi$, so $36h = 144$ and $h = 4$ cm (c) A: $2\pi(4)(9) = 72\pi$ cm². B: $2\pi(6)(4) = 48\pi$ cm². (d) Can B uses less: $72\pi - 48\pi = 24\pi$ cm²	4 pts: (a) Volume 144π cm³ 5 pts: (b) Sets up $\pi(6^2)h = 144\pi$ (3 pts); $h = 4$ (2 pts) 6 pts: (c) Label areas 72π and 48π (3 pts each) 5 pts: (d) Can B, by 24π cm²
S3 PYRAMID	An ice-cream cone has radius 3 cm and height 12 cm. A hemisphere of ice cream with the same radius sits on top. **(a)** Find the cone’s volume. **(b)** Find the hemisphere’s volume. **(c)** The cone is hollow. If all the ice cream melts into it, will it overflow? Explain with numbers. **(d)** Find the cone’s slant height in simplest radical form.	**Answer:** (a) 36π cm³ (b) 18π cm³ (c) No; 18π is less than 36π. (d) $3\sqrt{17}$ cm (a) $V = \frac{1}{3}\pi(3^2)(12) = 36\pi$ cm³ (b) $V = \frac{1}{2} \cdot \frac{4}{3}\pi(3^3) = \frac{1}{2} \cdot 36\pi = 18\pi$ cm³ (c) Melted ice cream keeps its volume, 18π cm³. The cone holds 36π cm³, so it fits and only fills half. (d) Slant height $= \sqrt{(3^2 + 12^2)} = \sqrt{153} = \sqrt{(9 \cdot 17)} = 3\sqrt{17}$ cm	4 pts: (a) Cone volume 36π cm³ 5 pts: (b) Half of the sphere volume: 18π cm³ 6 pts: (c) Compares 18π to 36π (3 pts); concludes it fits because volume stays the same (3 pts) 5 pts: (d) $\sqrt{153} = 3\sqrt{17}$ cm (2 pts setup, 3 pts simplest radical form)
S4 MODEL	A pool’s inside is a rectangular prism 12 m long, 6 m wide and 1.5 m deep. **(a)** Find its volume in m³ and in liters (1 m³ = 1000 L). **(b)** A pump adds 450 L per minute. How many hours to fill it? **(c)** The floor and four walls are tiled with 0.5 m by 0.5 m tiles. How many tiles are needed (no waste)?	**Answer:** (a) 108 m³ = 108,000 L (b) 4 hours (c) 504 tiles (a) $V = 12 \times 6 \times 1.5 = 108$ m³ = 108,000 L (b) $\frac{108{,}000}{450} = 240$ minutes = 4 hours (c) Floor: $12 \times 6 = 72$ m². Long walls: $2(12 \times 1.5) = 36$ m². Short walls: $2(6 \times 1.5) = 18$ m². Total = 126 m².	2 pts: (a) Volume 108 m³ 3 pts: (a) 108,000 L 3 pts: (b) 240 minutes 3 pts: (b) 4 hours 3 pts: (c) Floor 72 m² 3 pts: (c) Four walls 54 m², total 126 m² 3 pts: (c) 504 tiles using tile area 0.25 m²

		Each tile is $0.5 \times 0.5 = 0.25$ m², so $\frac{126}{0.25} = 504$ tiles.	